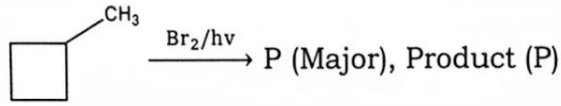
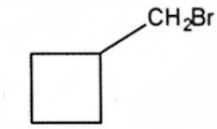
Haloalkane & Haloarenes

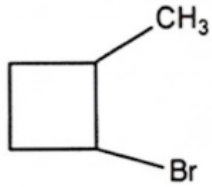
1. 2-Methylbutane on reacting with bromine in the presence of sunlight gives mainly
1) 1-bromo-2-methylbutane
2) 2-bromo-2-methylbutane
3) 2-bromo-3-methylbutane
4) 1-bromo-3-methylbutane

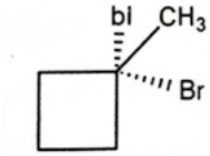
2. On halogenation, an alkane gives only one monohalogenated product. The alkane may be:
1) 2-methylbutane
2) 2,2-dimethylpropane
3) Cyclopentane
4) Both (b) and (c)

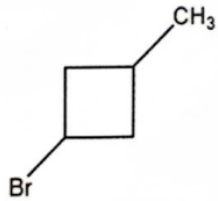


3. is

1) 

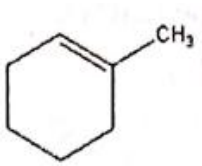
2) 

3) 

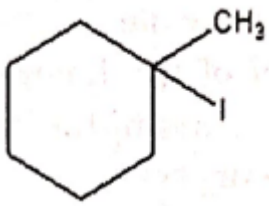
4) 

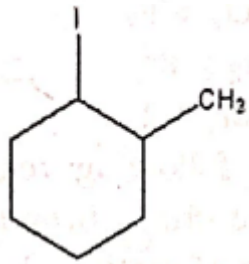
4. Which one of the following is gem dihalide?
1) $CH_3CH_2Br_2$
2) $CH_2Br - CH_2Br$
3) $CH_3CHBrCH_2Br$
4) $CH_3CHBrCH_2BH_2Br$

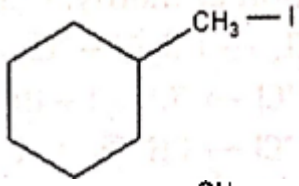
5. When cis-but-2-ene is treated with Br_2 in CCl_4 medium the product formed will be
1) (2R, 3S) dibromobutane
2) (2R, 3R) dibromobutane
3) (2S, 3S) dibromobutane
4) Mixture of (2R, 3R) and (2S, 3S) dibromobutane

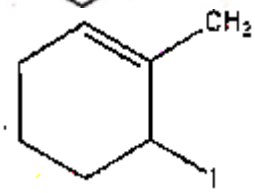
6. 

+ HI ⟶ major product is

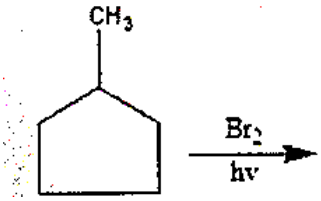
1) 

2) 

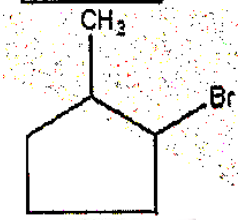
3) 

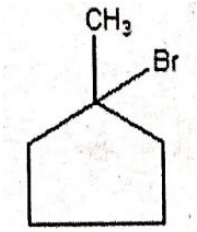
4) 

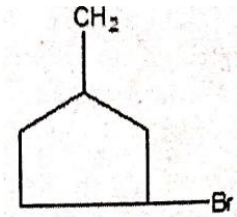
7. In the following reaction, the major product is:



1)

2) 

3) 

4) 

8. Initial attack in the reaction between 1- butene and bromine will be:

1) Bromonium ion
2) Bromide ion
3) Bromine atom
4) Bromine molecule

9. The reactivity of the halogens towards methane decreases in the order:

1) $F_2 > Cl_2 > Br_2 > I_2$
2) $I_2 > Br_2 > Cl_2 > F_2$
3) $F_2 > Cl_2 > I_2 > Br_2$
4) $Cl_2 > F_2 > Br_2 > I_2$

10. Which of the following reaction pairs constitutes the chain propagation step in chlorination of methyl chloride:

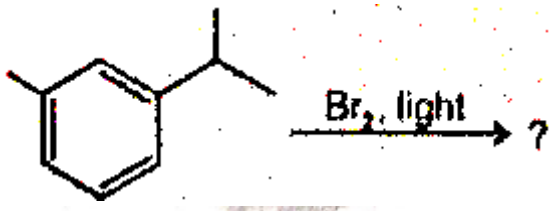
1) $^{\bullet}CH_3 + Cl_2 \rightarrow CH_3Cl + ^{\bullet}Cl$
$CH_3Cl + ^{\bullet}Cl \rightarrow ^{\bullet}CH_2Cl + HCl$

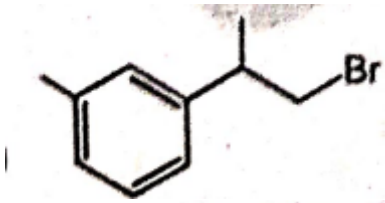
2) $CH_3Cl + ^{\bullet}Cl \rightarrow CH_2Cl_2 + ^{\bullet}H$
$^{\bullet}CH_2Cl + Cl_2 \rightarrow CH_2Cl_2 + ^{\bullet}Cl$

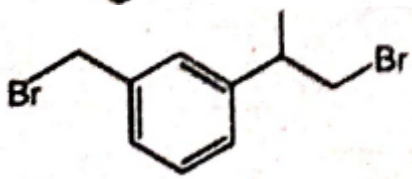
3) $CH_3Cl + ^{\bullet}Cl \rightarrow ^{\bullet}CH_2Cl + HCl$
$^{\bullet}CH_2Cl + Cl_2 \rightarrow CH_2Cl_2 + ^{\bullet}Cl$

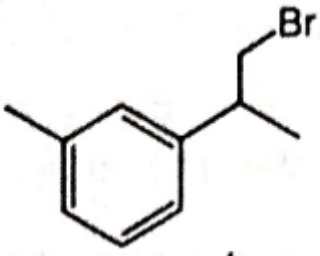
4) $^{\bullet}CH_2Cl + ^{\bullet}CH_2Cl \rightarrow CH_2Cl - CH_2Cl$
$^{\bullet}CH_2Cl + ^{\bullet}Cl \rightarrow CH_2Cl_2$

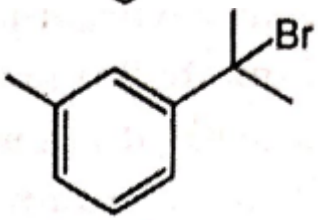
11. What is the major product of the following reaction?



1) 

2) 

3) 

4) 

12. Conversion of CH_4 to CH_3Cl is an example of which of the following reaction

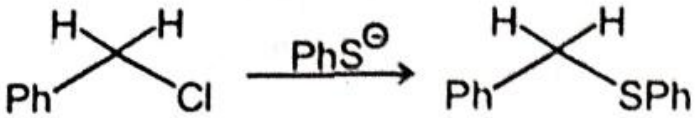
1) Electrophilic substitution
2) Free radical addition
3) Nucleophilic substitution
4) Free radical substitution

13. One mole of a symmetrical alkene on ozonolysis gives two moles of an aldehyde having a molecular mass of 44 u. The alkene is

1) 2-butene
2) Ethene
3) Propene
4) 1-butene

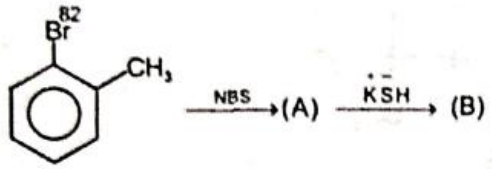
14. n-Pentane $\xrightarrow[h\nu]{Cl_2} (A)$ Mono-chloro product (including stereoisomers) are:

1) 2
2) 3
3) 4
4) 5

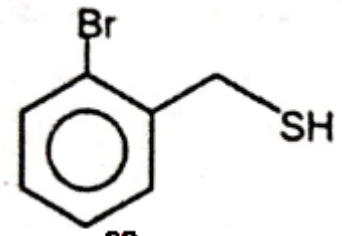
15. 

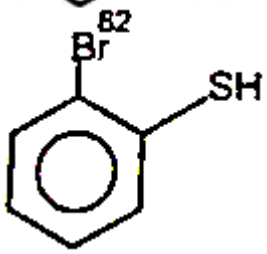
Given reaction is an example of

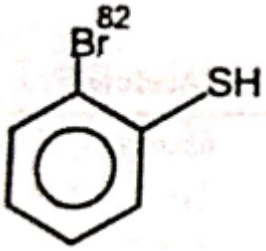
1) Electrophilic substitution
2) Electrophilic addition
3) Nucleophilic substitution
4) Free radical substitution

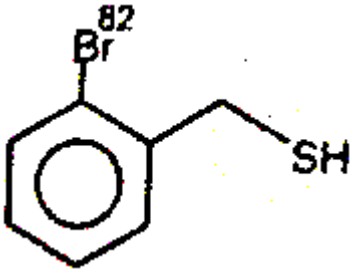
16. 

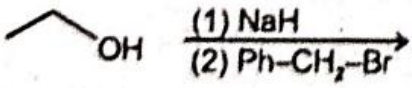
Product (B) is

1) 

2) 

3) 

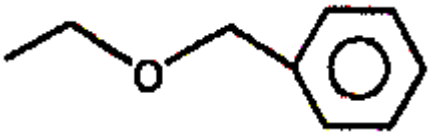
4) 



17. What will be the final product

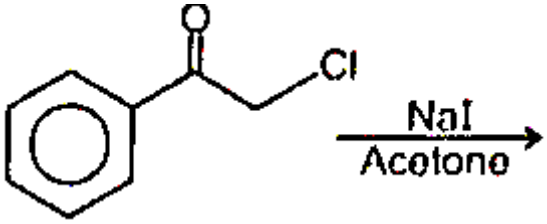
1)

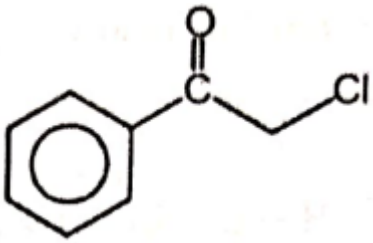
2) $H_2C = CH_2$

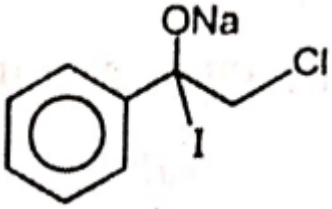
3) 

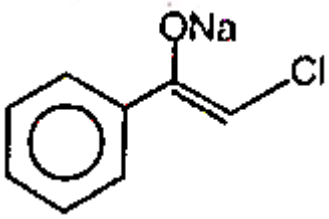
4) $Ph - CH_2 - CH_2 - CH_3$

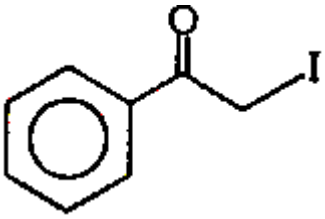
18. What will be the final product?



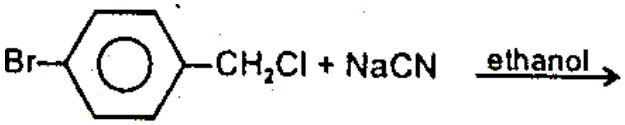
1) 

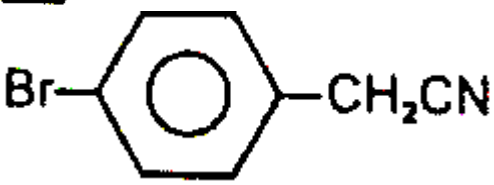
2) 

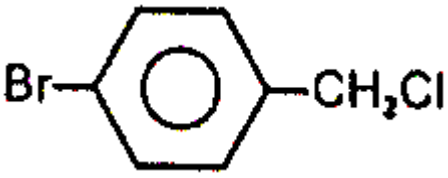
3) 

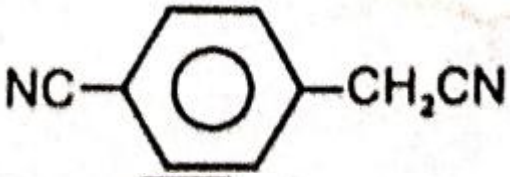
4) 

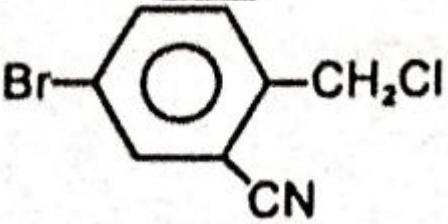
19. The product obtained from the reaction is:



1) 

2) 

3) 

4) 

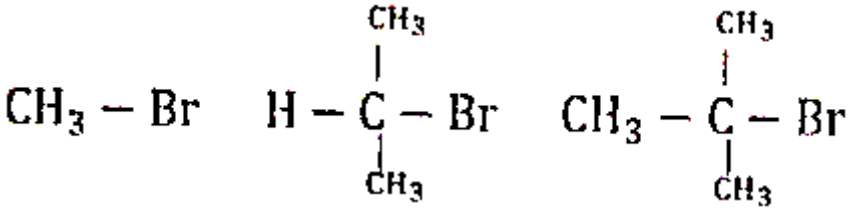
20. Correct nucleophilicity order will be

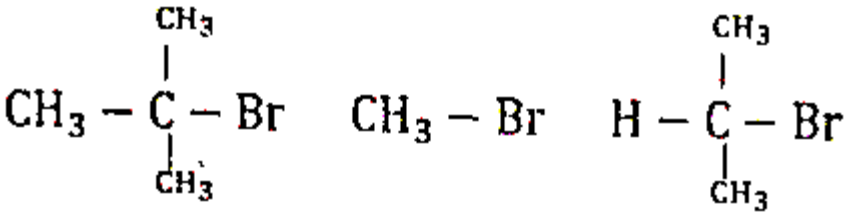
$(1)\, ph-S^{-} \qquad (2)\, HO^{-} \qquad (3)\, Ph-O^{-} \qquad (4)\, HS^{-}$

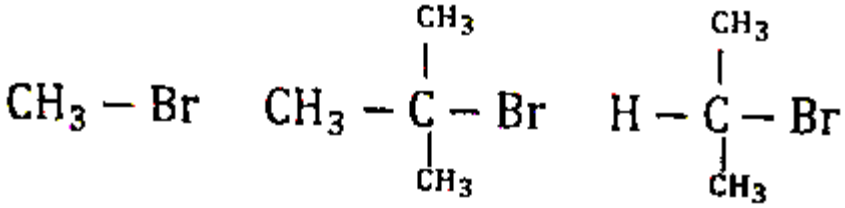
1) $2>3>1>4$
2) $4>1>2>3$

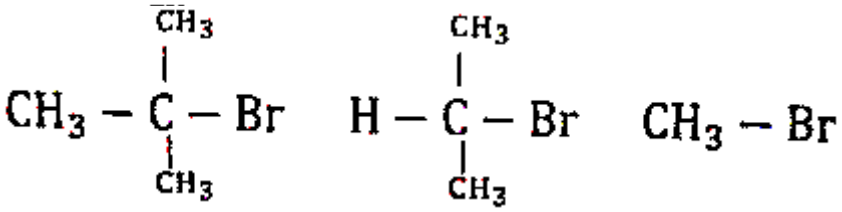
3) $1 > 4 > 3 > 2$
4) $1 > 3 > 4 > 2$

21. Choose the order that has the following alkyl bromides correctly arranged with respect to increasing reactivity in a SN_2 reaction

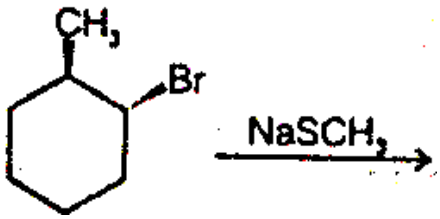
1) 

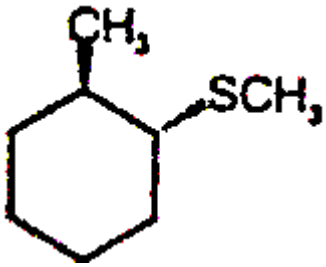
2) 

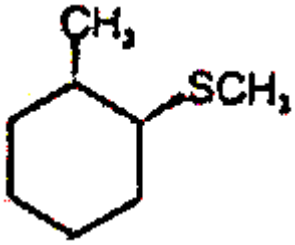
3) 

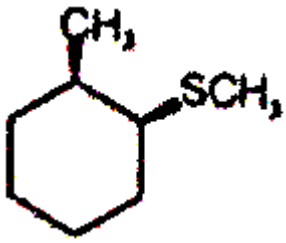
4) 

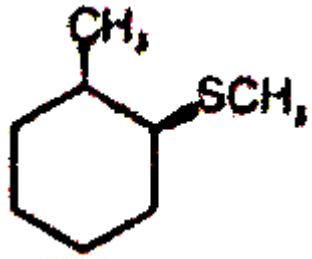
22. What product(s) would you expect from the following reaction?

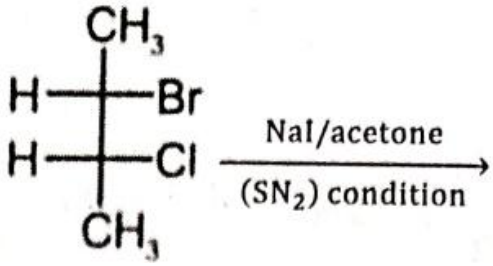


1) 

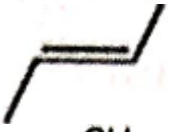
2) 

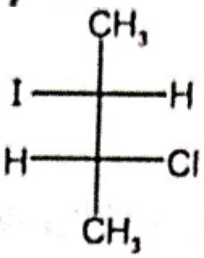
3) 

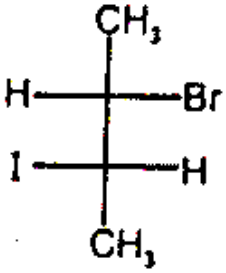
4) 

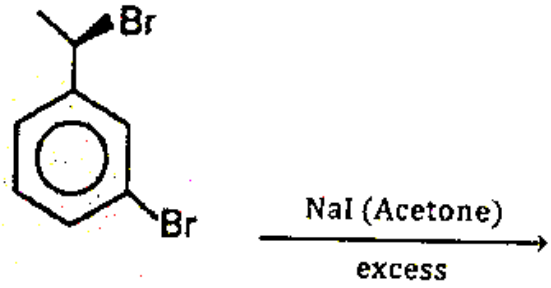
23. 

1)

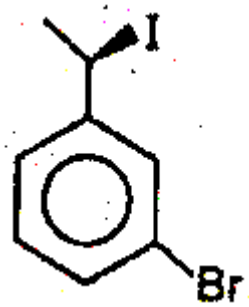
2) 

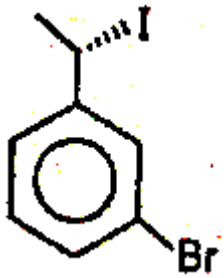
3) 

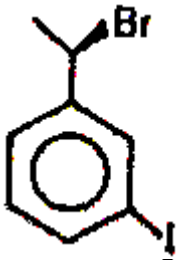
4) 

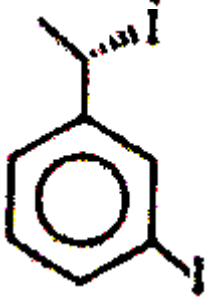
24. 

P, P is:

1) 

2) 

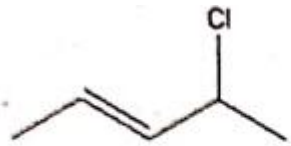
3) 

4) 

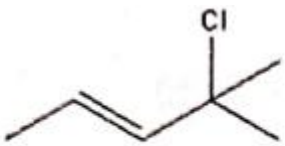
25. Geometry of reaction intermediate in SN_1 reaction is

1) Tetrahedral
2) Trigonal Planar
3) Triangular bipyramidal
4) None of these

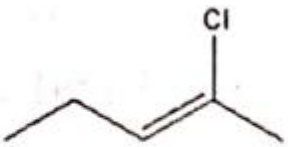
26. Which of the following will give two different alcohols upon SN1 hydrolysis? (including stereo)



(1)



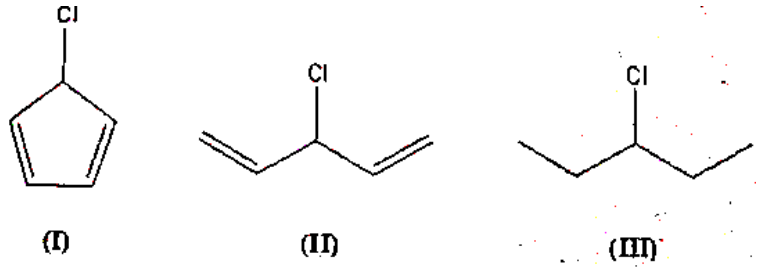
(2)



(3)

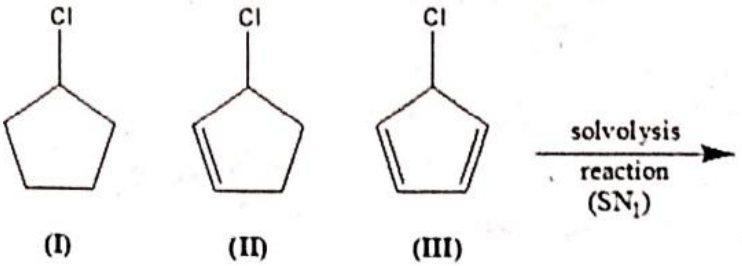
1) 1
2) 2
3) 3
4) Both 1 and 2

27. Arrange the following compounds in order of increasing SN_1 reactivity.



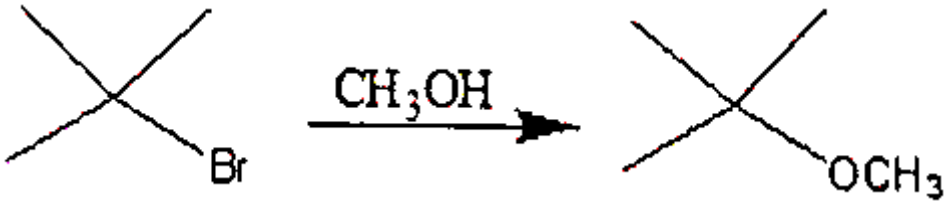
1) I < II > III
2) I < III > II
3) II < I > III
4) II < III < I

28. Choose the order that has the following compounds correctly ordered with respect to increasing reactivity with respect to a solvolysis (SN_1) reaction.



1) $I < II < III$
2) $I < III < II$
3) $II < III < I$
4) $III < I < II$

29. Which of the following would be true for the reaction shown?



1) The rate of the reaction depends only on the alkyl bromide concentration.
2) The rate of the reaction depends only on the methanol concentration.
3) The rate of the reaction depends on both the alkyl halide concentration and the methanol concentration.
4) The rate of the reaction depends on the concentration of neither reactant.

30. Which of the following is a **geminal dihalide**?

1) CH_2Cl-CH_2Cl
2) CH_3CHCl_2
3) CH_2Cl_2
4) $CCl_2=CH_2$

Answer Key

1	2	**16**	4
2	4	**17**	3
3	3	**18**	4
4	1	**19**	1
5	4	**20**	2
6	1	**21**	4
7	3	**22**	3
8	1	**23**	3
9	1	**24**	2
10	3	**25**	2
11	2	**26**	2
12	4	**27**	2
13	4	**28**	4
14	1	**29**	1
15	3	**30**	2